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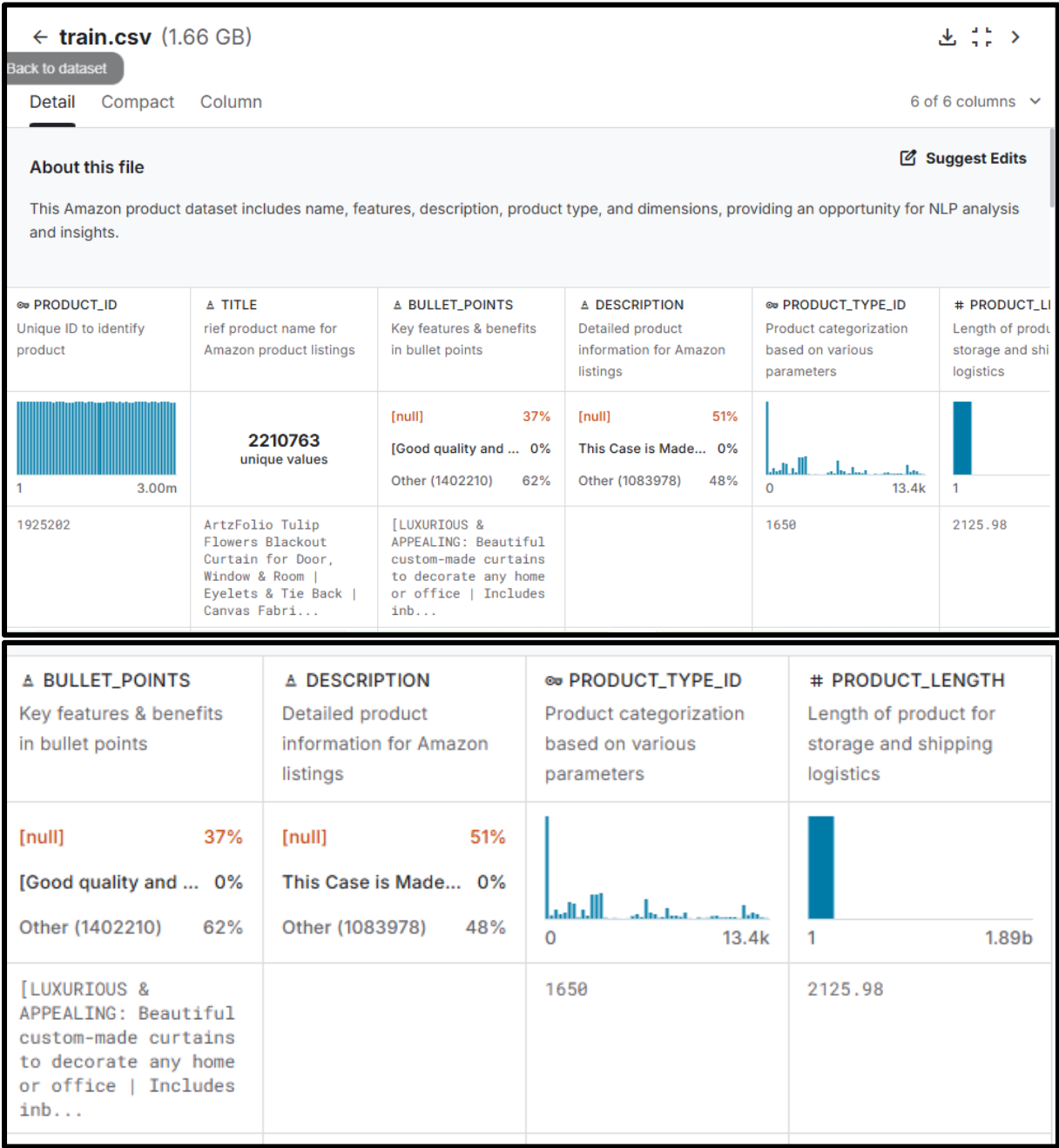
Name: Akshay Pande

DIV: SE2

PRN: 202501100130

Dataset: Amazon Product Dataset

Dataset Photo :



Dataset Columns

- PRODUCT_ID
- TITLE
- BULLET_POINTS
- DESCRIPTION
- PRODUCT_TYPE_ID
- PRODUCT_LENGTH

Dataset Assumption:

```
import pandas as pd
import numpy as np

df = pd.read_csv("train.csv")
```

Problems:

1. How many products are in the dataset?

```
df.shape[0]
```

2. Count missing values in each column

```
df.isnull().sum()
```

3. What percentage of products have missing descriptions?

```
(df['DESCRIPTION'].isnull().mean()) * 100
```

4. Find all products with missing bullet points

```
df[df['BULLET_POINTS'].isnull()]
```

5. What is the average product length?

```
df['PRODUCT_LENGTH'].mean()
```

6. What is the maximum and minimum product length?

```
df['PRODUCT_LENGTH'].agg(['min', 'max'])
```

7. Find products with unusually large length (outliers)

```
Q1 = df['PRODUCT_LENGTH'].quantile(0.25)
Q3 = df['PRODUCT_LENGTH'].quantile(0.75)
IQR = Q3 - Q1
```

```
df[(df['PRODUCT_LENGTH'] > Q3 + 1.5*IQR)]
```

8. Count number of unique product types

```
df['PRODUCT_TYPE_ID'].nunique()
```

9. Find top 10 most common product types

```
df['PRODUCT_TYPE_ID'].value_counts().head(10)
```

10. Find number of products per product type

```
df.groupby('PRODUCT_TYPE_ID')['PRODUCT_ID'].count()
```

11. Find products with longest titles

```
df['title_length'] = df['TITLE'].str.len()  
df.nlargest(5, 'title_length')[['TITLE',  
'title_length']]
```

12. Find products with shortest titles

```
df.nsmallest(5, 'title_length')[['TITLE',  
'title_length']]
```

13. Count number of words in description

```
df['desc_word_count'] =  
df['DESCRIPTION'].fillna('').apply(lambda x:  
len(x.split()))
```

14. Find products with very short descriptions (< 20 words)

```
df[df['desc_word_count'] < 20]
```

15. Find products with very long descriptions (> 100 words)

```
df[df['desc_word_count'] > 100]
```

16. Identify products with keyword "cotton" in description

```
df[df['DESCRIPTION'].str.contains('cotton',  
case=False, na=False)]
```

17. Count how many products mention "quality" in bullet points

```
df['BULLET_POINTS'].str.contains('quality',  
case=False, na=False).sum()
```

18. Normalize product length using NumPy

```
df['length_norm'] = (df['PRODUCT_LENGTH'] -  
np.min(df['PRODUCT_LENGTH'])) /  
(np.max(df['PRODUCT_LENGTH']) -  
np.min(df['PRODUCT_LENGTH']))
```

19. Find correlation between title length and description length

```
df['desc_length'] =  
df['DESCRIPTION'].fillna('').str.len()  
df[['title_length', 'desc_length']].corr()
```

20. Find duplicate product titles

```
df[df['TITLE'].duplicated()]
```